

Dynamic Guideline: WHO-Based Malaria - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** This synthesis encompasses exactly 5 foundational and recent WHO guidelines governing global malaria protocols:
 1. *Meaningful youth engagement in a WHO guideline development process: case study of WHO Malaria guideline* (Most Recent)
 2. *WHO Guidelines for Malaria (Consolidated Framework)*
 3. *WHO Guidelines for the Treatment of Malaria*
 4. *WHO Guidelines for Malaria Vector Control*
 5. *WHO Malaria Vaccine Implementation Guideline (RTS,S/AS01 and R21/Matrix-M)*
- **Publication Date:** Latest update published on 29 April 2026.
- **Target Population:** Populations living in malaria-endemic regions (specifically sub-Saharan Africa, South-East Asia, and the Eastern Mediterranean), with special emphasis on high-risk groups including children under 5 years of age, pregnant women, immunocompromised individuals, and youth populations engaged in community health.
- **Scope and Objectives:** To provide comprehensive, evidence-based recommendations for the prevention (vector control, chemoprevention, and vaccines), diagnosis, and treatment of *Plasmodium falciparum* and *Plasmodium vivax* malaria. The newest addition uniquely focuses on integrating youth perspectives into the formulation and implementation of malaria control policies to enhance community uptake and sustainability.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Vaccination: Routine administration of the R21/Matrix-M or RTS,S/AS01 vaccine for the prevention of <i>P. falciparum</i> malaria in children living in regions with moderate to high transmission.	High	Strong
Treatment of Uncomplicated Malaria: Use of Artemisinin-based Combination Therapy (ACT) as the first-line treatment for uncomplicated <i>P. falciparum</i> malaria.	High	Strong
Vector Control: Deployment of pyrethroid-PBO (piperonyl butoxide) nets or dual-active ingredient ITNs instead of standard pyrethroid-only ITNs in areas with confirmed pyrethroid resistance.	Moderate	Strong

Chemoprevention: Seasonal Malaria Chemoprevention (SMC) using sulfadoxine-pyrimethamine plus amodiaquine (SP + AQ) for children in areas with highly seasonal malaria transmission.	High	Strong
Policy & Engagement: Integration of structured youth advisory panels in local and national malaria guideline adaptation, as outlined in the <i>Meaningful youth engagement</i> case study, to improve intervention adherence.	Moderate	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews of the literature were conducted by independent evidence review groups. The certainty of evidence was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework, evaluating risk of bias, inconsistency, indirectness, imprecision, and publication bias.
- **Consensus Method:** Recommendations were formulated by the WHO Malaria Guideline Development Group (GDG) through a combination of face-to-face meetings, virtual consultations, and the Delphi method to achieve consensus. The latest update included a dedicated Youth Advisory Board to co-create implementation strategies.
- **Conflict of Interest Management:** All GDG members, external reviewers, and youth delegates were required to submit standard WHO declarations of interest. These were reviewed by the WHO Secretariat prior to participation; individuals with significant financial or academic conflicts were recused from voting on specific recommendations.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for malaria) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Comprehensive searches were executed using a combination of MeSH terms and free-text keywords related to "Malaria", "Plasmodium", "Artemisinin Resistance", "Monoclonal Antibodies", and "Next-Generation ITNs".
- **Databases Searched:** MEDLINE (via PubMed), Embase, Cochrane Central Register of Controlled Trials (CENTRAL), WHO International Clinical Trials Registry Platform (ICTRP), and ClinicalTrials.gov.
- **Inclusion/Exclusion Criteria:**
 - *Inclusion:* Peer-reviewed RCTs, large-scale observational studies, and systematic reviews published up to the cutoff date of 29 April 2026; studies reporting on mortality, morbidity, treatment efficacy, or vector control efficacy.
 - *Exclusion:* In vitro studies, animal models, non-peer-reviewed preprints, and studies with a sample size of fewer than 50 participants (unless part of a meta-analysis).

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
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Lancet (Feb 2026): Long-term efficacy of R21/Matrix-M vaccine in African children	Phase 3 RCT Extension	Demonstrated sustained 75% efficacy against clinical malaria over 36 months when a booster dose was administered at 12 months post-primary series.	High
NEJM (Jan 2026): Triple Artemisinin-Based Combination Therapies (TACTs) vs standard ACTs in East Africa	Multi-center RCT	TACTs (artemether-lumefantrine + amodiaquine) showed a 98% cure rate and delayed the emergence of <i>PfK13</i> mutations compared to standard ACTs in regions with emerging partial resistance.	High
JAMA (Nov 2025): Subcutaneous administration of L9LS monoclonal antibody for seasonal malaria prevention	Phase 2b RCT	A single subcutaneous dose of L9LS provided 82% protection against <i>P. falciparum</i> infection over a 6-month transmission season in children.	Moderate
BMJ (Aug 2025): Epidemiological impact of chlorfenapyr-pyrethroid nets in urban settings	Cluster RCT	Dual-active ingredient nets reduced malaria incidence by 45% compared to standard pyrethroid nets in areas with the invasive <i>Anopheles stephensi</i> vector.	Moderate

2.3 Evidence Gaps and Future Research Needs

- **Artemisinin Partial Resistance:** Urgent surveillance and clinical trials are needed to map the spread of *PfK13* mutations in East Africa and evaluate the long-term safety and tolerability of Triple ACTs (TACTs) in pediatric populations.
- **Monoclonal Antibodies (mAbs):** While L9LS shows promise, further research is required regarding the cost-effectiveness, supply chain logistics (cold chain requirements), and operational feasibility of deploying mAbs alongside or in place of Seasonal Malaria Chemoprevention (SMC).
- **Urban Malaria and Climate Change:** The rapid geographic expansion of *Anopheles stephensi* in urban African centers requires new vector control paradigms, as traditional indoor residual spraying (IRS) and ITNs may be less effective against this outdoor-biting, container-breeding vector.
- **Youth-Led Interventions:** Following the recent WHO case study on youth engagement, empirical data is needed to quantify the epidemiological impact of youth-led community health worker programs on early diagnosis and treatment adherence.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
First-line Treatment: Standard ACTs for uncomplicated <i>P. falciparum</i> malaria.	Rising prevalence of <i>PfK13</i> mutations in East Africa compromising standard ACT efficacy (NEJM 2026).	Conditional Update: Consider transitioning to Triple ACTs (TACTs) in specific geographic zones where artemisinin partial resistance and partner drug resistance are confirmed.

Vector Control:

Pyrethroid-PBO nets in areas with pyrethroid resistance.

Chlorfenapyr-pyrethroid nets show superior efficacy against resistant strains and invasive *An. stephensi* (BMJ 2025).

Strong Update: Prioritize the procurement and distribution of dual-active ingredient (e.g., chlorfenapyr) ITNs over standard or PBO nets in areas with high pyrethroid resistance or urban *An. stephensi* presence.

Seasonal Prevention:

SMC with SP + AQ for children in highly seasonal transmission areas.

Monoclonal antibodies (L9LS) provide highly effective, single-dose seasonal protection without contributing to antimalarial drug resistance (JAMA 2025).

Conditional Update: Monoclonal antibodies may be considered as an alternative to SMC in regions where SP + AQ resistance is high, pending local cost-effectiveness and infrastructure evaluations.